A REMARK ON OPENAI FAMILY 170: THE CASE $r(4,t)$

[AUTHOR NAME]

Abstract. OpenAI's manuscripts in family 170 of `github.com/openai/math` state the sharp logarithmic exponent $r(s,t) = t^{s-1}/(\log t)^{s-2+o(1)}$ only for $s \geq 5$ ($s = 5$ in a companion paper, $s \geq 6$ in the general paper, whose construction theorem is stated for projective dimension $d \geq 5$). We observe that the accompanying Lean formalization already proves the prime-field construction for every $d \geq 3$, and that the library's lower-bound transfer and upper bound need only $d \geq 1$ and $s \geq 3$. Instantiating at $d = 3$ gives, with a ten-line Lean file whose axioms are only `propext`, `Classical.choice`, `Quot.sound`,

$$\frac{t^3}{(\log t)^{2+\varepsilon}} \leq r(4,t) \leq C\frac{t^3}{(\log t)^2} \qquad (t \geq t_0(\varepsilon)),$$

so $r(4,t) = t^3/(\log t)^{2+o(1)}$. The previous best lower bound we are aware of is $\Omega(t^3/(\log t)^4)$ (Mattheus–Verstraëte). No new mathematics is claimed: the construction and every proof are OpenAI's.

1. Statement

Let $r(s,t)$ be the least N such that every simple graph on N vertices has a clique of order s or an independent set of order t.

Theorem 1. *There is $C > 0$ such that for every $\varepsilon > 0$ and all sufficiently large integers t,*

$$\frac{t^3}{(\log t)^{2+\varepsilon}} \leq r(4,t) \leq C\frac{t^3}{(\log t)^2}, \qquad \textit{hence} \qquad \lim_{t\to\infty} \frac{3\log t - \log r(4,t)}{\log\log t} = 2.$$

This is exactly the statement `MainBounds 4 ∧ MainLimit 4` in the notation of the comparator file `lean/ComparatorChallenges/SharpLogRamsey.lean`, whose public theorem is stated for $6 \leq s$.

Before / after.

	lower bound for $r(4,t)$	source
Before	$c\,t^3/(\log t)^4$	Mattheus–Verstraëte [4]; Bradač [2] at $s = 4$
After	$t^3/(\log t)^{2+\varepsilon}$, every $\varepsilon > 0$	OpenAI 170 Lean library at $d = 3$ (this remark)
Upper	$C\,t^3/(\log t)^2$	Ajtai–Komlós–Szemerédi [1]; see also [3]; also in the 170 library

2. Where the statement comes from

In the general manuscript (*Sharp Logarithmic Exponents for Fixed Off-Diagonal Ramsey Numbers*, Theorem 1.2) the construction reads: for $d \geq 5$, $0 < \eta < 1/10$ and every sufficiently large prime q, with $\sigma = \log q$, there is a K_{d+1}-free graph on $\lfloor q^d\sigma \rfloor$ vertices with independence number $< \lfloor q\sigma^{1+\eta} \rfloor$ (the random ordered incident-flag graph of $\mathrm{PG}(d,q)$). In Lean this is the proposition `LowerTransfer.PrimeConstruction d`. The library contains

Date: October 9, 2026.

```
theorem actual_prime_construction (i : ℕ) : PrimeConstruction (i+3)
theorem prime_construction (d : ℕ) (hd : 5 ≤ d) : PrimeConstruction d
theorem all_epsilon_lower (d : ℕ) (hd : 1≤d) (hprime : PrimeConstruction d)
    (ε : ℝ) (hε : 0<ε) : ∀ᶠ t : ℕ in atTop,
      (t:ℝ)^d/(Real.log t)^(((d-1:ℕ):ℝ)+ε) ≤ (ramsey (d+1) t :ℝ)
theorem Upper.ramsey_upper {s : ℕ} (hs : 3 ≤ s) : ∃ C : ℝ, 0 < C ∧ ∃ t₀ : ℕ,
    ∀ t : ℕ, t₀ ≤ t → (ramsey s t : ℝ) ≤ C*(t:ℝ)^(s-1)/(Real.log (t:ℝ))^(s-2)
```

(`Construction/PrimeConstruction.lean`, `Bounds/Lower.lean`, `Bounds/SamplingParameter.lean`). The restriction $d \geq 5$ appears only in the wrapper `prime_construction`, which is itself proved from `actual_prime_construction (d-3)`; the formal proof of `actual_prime_construction` is uniform in $i \geq 0$, i.e. in $d \geq 3$ (vector dimension $i + 4$). The public theorem `main_of_primeConstruction` assumes $6 \leq s$ but its proof uses only $3 \leq s$ (upper bound) and $1 \leq s - 1$ (transfer).

The manuscript itself is consistent with this: its geometric core is carried out in projective dimensions 2 and 3, and higher dimensions $j \geq 4$ are reduced to these by repeated projection (Section 2, Proposition "Repeated projection"). Neither manuscript gives a reason for excluding $s = 4$; the companion $r(5, t)$ paper remarks that one of its counting bounds "also covers dimension three". We did not re-audit the manuscript at $d = 3$; the evidence offered here is the machine-checked proof.

3. The Lean certificate

```
import OAI.Combinatorics.SharpRamsey.Main
open OAI.SharpLogRamsey

theorem r4_lower (ε : ℝ) (hε : 0 < ε) :
    ∀ᶠ t : ℕ in Filter.atTop,
      (t:ℝ)^3/(Real.log t)^(((3-1:ℕ):ℝ)+ε) ≤ (ramsey (3+1) t : ℝ) :=
  LowerTransfer.all_epsilon_lower 3 (by norm_num)
    (LowerTransfer.actual_prime_construction 0) ε hε

theorem main_four : MainBounds 4 ∧ MainLimit 4 := by
  have hb : MainBounds 4 := by
    obtain ⟨C, hC, t₀, ht₀⟩ := Upper.ramsey_upper (s := 4) (by norm_num)
    refine ⟨C, hC, fun ε hε => ?_⟩
    have hl := LowerTransfer.all_epsilon_lower 3 (by norm_num)
      (LowerTransfer.actual_prime_construction 0) ε hε
    obtain ⟨t₁, ht₁⟩ := Filter.eventually_atTop.mp hl
    exact ⟨max t₀ t₁, fun t ht => ⟨ht₁ t ((le_max_right _ _).trans ht),
      ht₀ t ((le_max_left _ _).trans ht)⟩⟩
  exact ⟨hb, mainLimit_of_mainBounds 4 hb⟩
```

Here `ramsey s t` is $\inf\{N : \ \forall G \text{ on } \mathrm{Fin}\, N,\ G \text{ has an } s\text{-clique or } G^c \text{ has a } t\text{-clique}\}$, which is $r(s, t)$; the definitions in `Model.lean` are textually identical to those of the comparator file. Pinned to `openai/math` commit `fd4aeeb2ee4f` (the `SharpRamsey` directory is unchanged since the initial commit `adc7f1241b42`), Lean `v4.34.1`, both theorems elaborate and `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. The `SharpRamsey` sources contain no `sorry`, `axiom`, `native_decide`, `implemented_by`, `unsafe` or `set_option`, and import nothing from `OAI` outside that directory.

Remark 2. The same observation at $i = 1$ ($d = 4$) recovers the $r(5, t)$ statement from the general library, independently of the separate `RamseyFive` formalization. We do not pursue this.

Scope

Nothing here is claimed as new mathematics; the construction, its proof and its formalization are OpenAI's. The remark only records that the existing formal proof covers $s = 4$, which the manuscripts and the comparator statement do not state. To our knowledge (search of arXiv, the web and the `openai/math` GitHub repository up to 2026-10-09) no earlier source states a lower bound for $r(4,t)$ with logarithmic power below 4.

Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

References

[1] M. Ajtai, J. Komlós, E. Szemerédi, A note on Ramsey numbers, *J. Combin. Theory Ser. A* **29** (1980), 354–360.
[2] D. Bradač, Off-diagonal Ramsey numbers, arXiv:2605.28793v3 (2026).
[3] Y. Li, C. C. Rousseau, W. Zang, Asymptotic upper bounds for Ramsey functions, *Graphs Combin.* **17** (2001), 123–128.
[4] S. Mattheus, J. Verstraëte, The asymptotics of $r(4,t)$, *Ann. of Math.* **199** (2024), 919–941.
[5] OpenAI, Sharp logarithmic exponents for fixed off-diagonal Ramsey numbers; The sharp logarithmic exponent of $r(5,t)$, OpenAI Math Release (2026), `https://github.com/openai/math`, family 170.